Emily Fang

Raleigh, NC | efang2@ncsu.edu | (919) 396-8937 | linkedin.com/in/ehfang | ehfang.github.io

Education

North Carolina State University, Raleigh, NC

Ph.D. in Industrial Engineering *(GPA: 3.94)* — *Aug 2024 - Present*

Human Factors Engineering, Human-Computer Interaction

B.S. in Industrial Engineering *(GPA: 3.77)* — *Aug 2019 - May 2024*

Minor in Supply Chain Management

Experience

North Carolina State University - Graduate Research Assistant — *May 2024 - Present*

AI-Assisted Augmented Reality Checklists for Occupational Hazard Recognition and Mitigation — *Aug 2025 - Oct 2026*

- Developed an augmented reality prototype in Unity (C#), deploying to Meta Quest 3 using the passthrough camera API to overlay hazard-mitigation checklists in the user's real environment.
- Integrated OpenAI GPT-4o via API with a gesture-driven Unity UI, sending Meta Quest 3 passthrough camera snapshots to generate context-specific hazard-mitigation checklists and parsing structured JSON into interactive canvas elements.

Short-Term Cognitive Effects of Different Video Game Tasks — *May 2024 - Present*

- Administered 2-hour in-lab experimental protocols with adult participants, conducting standardized cognitive assessments to measure selective attention, inhibitory control, working memory, and multitasking abilities pre- and post-intervention.
- Facilitated computer-based tasks, conducted semi-structured interviews, and collected behavioral performance, eye-tracking, and survey data (Qualtrics, PANAS, NASA-TLX) for quantitative and qualitative analysis.

Scale Cognition Advanced Learning Environments in Virtual Reality (SCALE-VR) — *Aug 2023 - Jul 2025*

- Collaborated with interdisciplinary teams across education, design, and engineering to assist in the iterative development of an NSF-funded VR application (Award #2055680) designed to teach students concepts of scale and numeracy.
- Conducted qualitative analysis on formative evaluation data through qualitative coding and Thematic Analysis, identifying insights to enhance usability and user experience.
- Trained an undergraduate researcher in thematic analysis over a semester; he subsequently coded qualitative data independently.

North Carolina State University - Undergraduate Research Assistant — *Aug 2021 - May 2024*

Augmented Reality for Engineering Education Enhancement (AREEA) — *Oct 2022 - May 2024*

- Collaborated with a Ph.D. researcher to conduct a mixed-methods evaluation of a game-based augmented reality application aimed at teaching statistics, engaging directly with participants to collect both qualitative and quantitative data.
- Facilitated data collection by administering validated questionnaires, conducting interviews, and guiding participants through evaluation tasks. Conducted qualitative and quantitative analysis to generate insights on usability and UX.

Virtual Instructor Application 2 (VIA 2) — *Aug 2021 - Oct 2022*

- Assisted a Ph.D. researcher to iteratively improve an augmented reality-based postural training tool for occupational safety, applying both qualitative and quantitative research methods to evaluate and refine usability and effectiveness.
- Utilized various research methodologies, including PSSUQ (Post-Study System Usability Questionnaire), Bipolar Laddering, and Semi-Structured Interviews to deliver actionable insights on tool usability and training outcomes.

Publications

- Cheng, F., **Fang, E.**, & Chen, K. B. (2026). Elicitation and Assessment of Hand-Based Gestures for Information Processing in Immersive Environments. ***Human Factors: The Journal of the Human Factors and Ergonomics Society***. https://doi.org/10.1177/00187208261480471
- **Fang, E.**, & Chen, K. B. (2026). Examining Short-Term Video Game Play on Subsequent Working Memory Performance. ***Proceedings of the Human Factors and Ergonomics Society Annual Meeting***. https://doi.org/10.1177/10711813261484466
- Chen, K. B., Alabi, C., Bailey, S. K. T., **Fang, E.**, Fussell, S. G., Harris, J., Akash, K., & Wu, L. (2026). Human Factors Extended Reality Showcase 2026. ***Proceedings of the Human Factors and Ergonomics Society Annual Meeting***. https://doi.org/10.1177/10711813261498754
- Perera, G. N., **Fang, E.**, Bottomley, L., Chen, K. B., & Ivy, J. (2025). Study Design and Assessment Framework for Testing Augmented Reality Tools in Engineering Education. ***2025 ASEE Annual Conference & Exposition***. https://peer.asee.org/57157 **[1st place winner for Best New Career Paper Award]**
- **Fang, E.**, Kulasingam, R., Cheng, F., Peterson, M., Delgado, C., & Chen, K. B. (2025). Examining User Interactions With Signaling Elements in a Virtual Reality Learning Application. ***Proceedings of the Human Factors and Ergonomics Society Annual Meeting***. https://doi.org/10.1177/10711813251360711
- Cheng, F., **Fang, E.**, & Chen, K. B. (2025). Mental Models of Gestural Interaction for Information Processing in Virtual Reality. ***Proceedings of the Human Factors and Ergonomics Society Annual Meeting***. https://doi.org/10.1177/10711813251357931
- **Fang, E.**, Sivaramakrishnan, A., & Chen, K. B. (2024). Negative Emotions From Virtual Reality Usage: A Preliminary Exploratory Study Using Online Forums. ***Proceedings of the Human Factors and Ergonomics Society Annual Meeting***. https://doi.org/10.1177/10711813241275079

- Chen, K., Perera, G., **Fang, E.**, & Chen, K. B. (2022). Augmented Reality (AR) Posture Training for Manual Material Handling: Iterative Design, Evaluation, and Recommendation for AR Interface. ***Proceedings of the Human Factors and Ergonomics Society Annual Meeting***. https://doi.org/10.1177/1071181322661093

Under Review

- Perera, G. N., **Fang, E.**, Chen, K., Ivy, J. S., & Chen, K. B. (under review). Worker-Centered Design and Evaluation of Augmented Reality Application for Posture Training. ***Ergonomics in Design***.

In Preparation

- Wu, L., **Fang, E.**, Cheng, F., & Chen, K. B. (in preparation). Balancing Usability and Theory: An Iterative Evaluation Framework for Virtual Learning Environments.

Presentations and Demonstrations

- Augmented Reality Hazard Detection: An AI-Assisted Approach to Construction Safety. XR Showcase demo, *Human Factors and Ergonomics Society Annual Meeting*, 2026. Co-presented with M. Malaek and M. Roy.
- Examining Short-Term Video Game Play on Subsequent Working Memory Performance. Poster presentation, *Human Factors and Ergonomics Society Annual Meeting*, 2026.
- Examining User Interactions With Signaling Elements in a Virtual Reality Learning Application. Lecture presentation (20 min), *Human Factors and Ergonomics Society Annual Meeting*, 2025.
- Negative Emotions From Virtual Reality Usage: A Preliminary Exploratory Study Using Online Forums. Lecture presentation (20 min), *Human Factors and Ergonomics Society Annual Meeting*, 2024.
- Stratify Subjective Numeric Ratings and Qualitative Feedback of Augmented Reality Ergonomic Posture Training Tool. Poster presentation, *IISE Applied Ergonomics Conference*, 2023.

Honors and Awards

- National Institute of Occupational Safety and Health Graduate Fellowship at NC State University, *2025 - Present*
- Provost's Doctoral Fellowship at NC State University, *2024 - 2025*
- Graduate Merit Award at NC State University, *2024 - 2025*
- Institute of Industrial and Systems Engineers Applied Ergonomics Conference Student of the Year Award, *2023*

Teaching Experience

- Grader, ISE 540: Human Factors in Systems Design, North Carolina State University, Spring 2026

Professional Service

- Reviewer, *Human Factors and Ergonomics Society Annual Meeting*, 2025 - 2026
- Session Chair, XR6: Understanding Human Performance and Physiological Responses in XR, Extended Reality Technical Group, *Human Factors and Ergonomics Society Annual Meetin*g, 2025

Outreach

- Led hands-on demonstrations of Scale Worlds, an NSF-funded VR learning environment, at the NC Science Museum and summer camps; assisted with demonstrations at a local underserved middle school, NC State Engineering Open House, and lab tours for visiting seminar speakers
- Counselor, Industrial and Systems Engineering Summer Camp for High School Students, The Engineering Place, North Carolina State University, Summers 2025 - 2026. Mentored about 25 high school campers on college, research, and ISE; designed interactive end-of-camp review games covering manufacturing and human factors tools.

Professional Memberships

- Institute of Industrial and Systems Engineers (IISE), 2022 - Present
- Human Factors and Ergonomics Society (HFES), 2022 - Present

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Skills

- Python, PyTorch, scikit-learn, C#, JavaScript, R, Unity3D, Meta XR SDK, OpenAI API, Streamlit, Gurobi, Git/GitHub
- Qualtrics, Dedoose, NVivo, Tableau, Figma, PsyToolkit, Pupil Labs (Pupil Core)
- Mixed Methods, Qualitative and Quantitative Analysis, Signal Detection Theory, Experimental Design, Eye Tracking & Pupillometry, Thematic Analysis, Usability Analysis, Formative Evaluation, Think-Aloud Protocols, User-Centered Design, Study Coordination, Technical Writing
- Experimental Paradigms: n-back, flanker, task switching
- Languages: English, Mandarin (native/bilingual)